

The Potential Outcomes Framework

Overview

[date]

- learn the notation of potential outcomes and define individual-level effects using potential outcomes
- formulate average causal effects using potential outcomes notation, including average treatment effects (ATEs), average treatment effects on the treated, and conditional ATEs
- understand common challenges in causal attribution using the potential outcomes framework

Readings

- https://mixtape.scunning.com/04-potential_outcomes#statistical-inference
- https://mixtape.scunning.com/04-potential_outcomes#physical-randomization
- https://mixtape.scunning.com/04-potential_outcomes#potential-outcomes
- https://mixtape.scunning.com/04-potential_outcomes#average-treatment-effect

Async

Setting the Stage: From Urn Models to Box Models

<https://www.degruyterbrill.com/document/doi/10.1515/jci-2023-0073/html>

Summary

- Q: how can we explain Neyman's 1923 ideas about causal inference in a way that students can actually understand?
- A: use simple physical models, especially:
 - urns
 - boxes
 - tickets
 - random draws
 - two possible outcomes per subject

	the article argues that Neyman's ideas are still underused in introductory statistics education, even though they are central to modern causal inference
Neyman's 1923 Framework	- in an experiment, each unit has potential outcomes - example: - a person in a medical trial has - one outcome if treated - one outcome if not treated - but we can only observe one of those outcomes - that is the core causal-inference problem - the missing outcome is called the counterfactual outcome - the authors explain that Neyman's setup treats the actual experimental units as the population of interest - the randomness comes from random treatment assignment - not from assuming the subjects were randomly sampled from some larger imaginary population
Potential Outcome	- a possible result for one unit under one treatment condition - for unit i: $y_i(1)$ = outcome if treated $y_i(0)$ = outcome if untreated / control - only one can be observed for each unit - the appendix uses exactly this setup: $y_i(1)$ and $y_i(0)$ are fixed quantities, while treatment assignment indicators are random

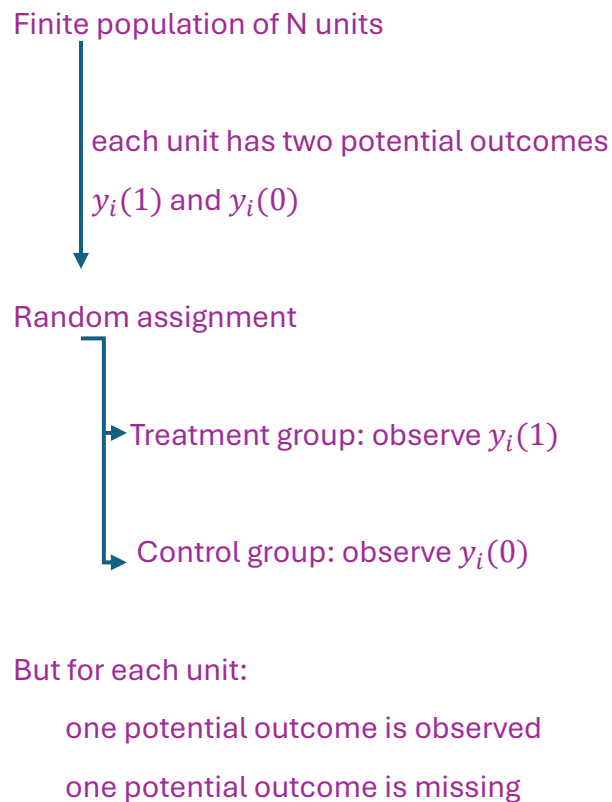
Average Treatment Affect (ATE)	• $\hat{y}(1)$ = average outcome if everyone were treated = • $\hat{y}(0)$ = average outcome if everyone were untreated • $ATE = \hat{y}(1) - \hat{y}(0)$ • $\widehat{ATE}$ = mean observed outcome in treatment group – mean observed outcome in control group
Randomization-Based Inference	• the uncertainty comes from which units were randomly assigned to which treatment • it does not require imagining that the experiment's subjects are a random sample from a huge external population • this is different from the common textbook approach, where two-sample tests are often justified by assuming two independent samples from two infinite or hypothetical populations
Conservative Standard Error	• a standard error is conservative when it is too large, or tends to overstate uncertainty • that is usually safer than being too small • the article emphasizes that the usual standard error in randomized experiments is often conservative because of two mistakes that partly cancel each other
Neyman's First Major Result	• difference in group means is an unbiased estimator of the average treatment effect • so, for a randomly assigned treatment $E[\widehat{ATE}] = ATE$ • so, the observed treatment-control difference is correct on average over repeated random assignments

	this works because the treatment group and control group are random samples from the finite set of experimental units
The Surprising Standard-Error Result	- the usual variance estimator for a difference in means is based on: $\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}$ - where: $s_1^2 = \text{sample variance in the treatment group}$ $s_0^2 = \text{sample variance in the control group}$ $n_1 = \text{treatment} - \text{group size}$ $n_0 = \text{control} - \text{group size}$ - traditional textbooks often justify this formula by pretending there are: - two independent samples - from two infinite populations - but, in Neyman's randomized experiment setup: - the population is finite - the treatment and control groups are not independent - if one subject is in treatment, that same subject cannot also be in control - so, the usual textbook justification is technically wrong for randomized experiments
The "Minor Miracle"	- Freedman, Pisani, and Purves explain the result using a box-of-tickets model - each ticket represents one subject - each ticket has two numbers $A = \text{outcome under treatment } A$ $B = \text{outcome under treatment } B$

	○ but, once a subject is assigned to one treatment, only one number is visible ○ the other is hidden ○ this model makes potential outcomes easier to see visually • the usual standard error makes two mistakes: <table border="1"> <thead> <tr> <th>Mistake</th><th>What it does</th></tr> </thead> <tbody> <tr> <td>treats sampling as if it were with replacement</td><td>makes the SE too big</td></tr> <tr> <td>treats the two sample means as independent</td><td>makes the SE smaller</td></tr> </tbody> </table> • the surprising result is that these two mistakes partly cancel • this is why the ordinary two-sample standard-error formula can still be usable for randomized experiments • Freedman, Pisani, and Purves describe this as a “minor miracle”	Mistake	What it does	treats sampling as if it were with replacement	makes the SE too big	treats the two sample means as independent	makes the SE smaller
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When the Usual Variance Estimator Works	• Case 1: Constant additive treatment effect ○ this means every unit is affected by the same amount ○ example: ▪ ‘everyone improves by exactly 5 points under treatment’ ○ then the usual variance estimator is unbiased ○ so, it estimates the variance correctly on average • Case 2: Heterogeneous treatment effects ○ this means treatment effects vary across units ○ example: ▪ some people improve a lot ▪ some improve a little						

	▪ some may get worse ○ then, the usual variance estimator is generally conservative ○ it tends to overestimate variance by an amount related to the finite-population variance of individual treatment effects: $\text{conservative bias} \approx \frac{\sigma_{\Delta}^2}{N}$ ○ where: σ_{Δ}^2 measures how much treatment effects vary across units N is the number of subjects in the finite population ○ if N is much larger than the treatment and control sample sizes, this bias becomes negligible
Important Warning About Pooled Variance	• the article also notes a technical caution: ○ Neyman did not study the pooled variance estimator ○ the pooled estimator can behave differently • if treatment and control group sizes are unequal and treatment effects are heterogeneous, the pooled estimator may be: ○ conservative ○ anticonservative • Anticonservative means it can underestimate uncertainty which is dangerous because it can make results look more precise than they really are
The Article's Broader Argument	• most introductory statistics textbooks still explain two-sample tests using independent samples from hypothetical populations • the authors think this misses an important opportunity

- in randomized experiments, we often do not need to imagine hypothetical superpopulations.
- we can instead say:
 - the experimental units are the population
 - random assignment creates the uncertainty
 - causal inference is hard because each unit has missing potential outcomes
- the authors do not claim randomization-based inference is always better than superpopulation inference
- their claim is narrower:
- Students should learn both frameworks.
 - randomization-based causal inference is currently underemphasized

**Conceptual
Diagram**

	Estimate: $\text{ATE_hat} = \text{treatment mean} - \text{control mean}$ Core result: $\text{ATE_hat is unbiased for ATE}$ Standard-error result: usual SE formula often works because two errors partly cancel
Most Important Takeaway	• Neyman's 1923 contribution is not just historical, it gives a clean way to understand randomized experiments: ○ causal uncertainty comes from random assignment ○ each unit has multiple potential outcomes ○ only one outcome is observed ○ the difference in means is unbiased for the average treatment effect ○ the usual standard error can be justified in a finite-population framework ○ but its behavior depends on treatment-effect heterogeneity • the authors' educational point is that these ideas do not have to be hidden behind advanced notation • they can be taught with boxes, tickets, diagrams, and careful explanation

Introduction to the Potential Outcomes Framework

Notation and Definitions

Treatment assignment:

- Binary treatment: $D_i \in \{0,1\}$
- Framework can be extended to continuous treatments (or doses)

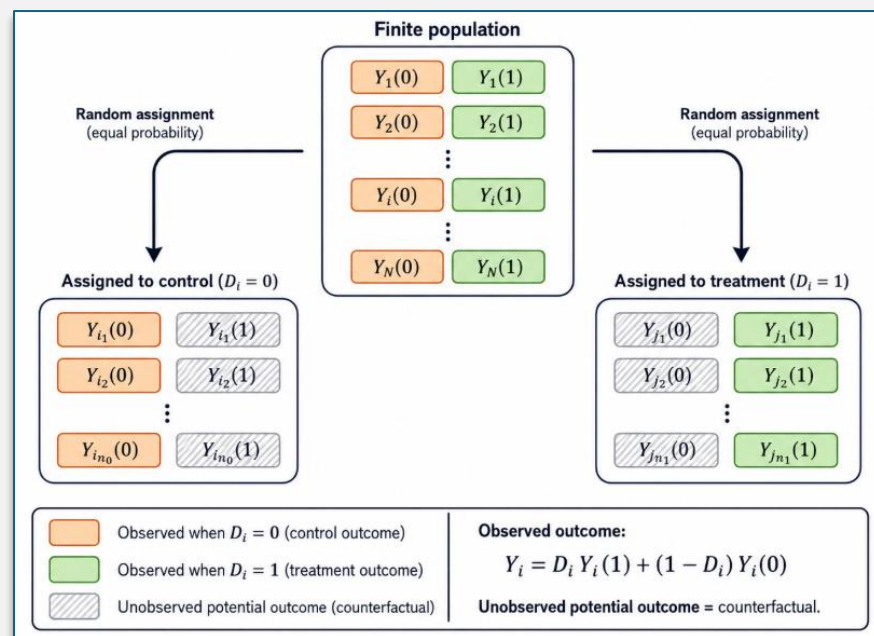
Potential outcome:

- $Y_i(1)$: outcome if unit i receives treatment
- $Y_i(0)$: outcome if unit i does not receive treatment

Observed outcome:

$$Y_i = D_i \cdot Y_i(1) + (1 - D_i) \cdot Y_i(0)$$

Neyman's Urn Model



Example: Get Out the Vote Campaign	Voter	Treatment (received GOTV call?)	$Y_i(1)$ (turnout with call)	$Y_i(0)$ (turnout without call)	Observed turnout (Y_i)
	Alice	1	1 (votes)	0 (doesn't vote)	1
	Bob	0	0 (doesn't vote)	0 (doesn't vote)	0
	Carol	1	1 (votes)	1 (votes anyway)	1
Individual Treatment Effects (ITEs)	individual treatment effect (ITE):				
	$\tau_i = Y_i(1) - Y_i(0)$				
	alternative definitions are possible:				
	$\frac{Y_i(1)}{Y_i(0)} \text{ (less common)}$ $F_{\{Y_i(1)\}^{-1}}(p) - F_{\{Y_i(0)\}^{-1}}(p) \text{ (quantile p effect)}$ the causal effect captures what would happen to the same individual under two alternative scenarios: treated vs. untreated counterfactual by definition!				